

Cybersecurity - Undergraduate Level Assessment 22

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

To verify a digital signature we need the

- A. Sender's Private key
- B. Sender's Public key
- C. Receiver's Private key
- D. Receiver's Public key

Question 2

(2 marks)

Which of the following deals with network intrusion detection and real-time traffic analysis?

- A. John the Ripper
- B. L0phtCrack
- C. Snort
- D. Nessus

Question 3

(2 marks)

The digest created by a hash function is normally called a

- A. Modification detection code (MDC)
- B. Modify authentication connection
- C. Message authentication control

D. Message authentication cipher

Question 4

(2 marks)

What does EXE do when the STP solver times out on a constraint query for a particular path?

- A.** Assume that the query is satisfiable and continue executing the path.
- B.** Assume that the query is not satisfiable and stop executing the path
- C.** Restart STP and retry the query, up to a limited number of retries.
- D.** Remove a subset of the constraints and retry the query.

Question 5

(2 marks)

Can a stream cipher have perfect secrecy?

- A.** Yes, if the PRG is really "secure"
- B.** No, there are no ciphers with perfect secrecy
- C.** Yes, every cipher has perfect secrecy
- D.** No, since the key is shorter than the message

True / False

Question 1

(2 marks)

A man-in-the-middle attack cannot affect the Diffie-Hellman method if the two parties are joined in a secure network.

True

False

Question 2

(2 marks)

The sub key length at each round of DES is 56 bits.

True

False

Question 3

(2 marks)

SQL queries based on user input are particularly vulnerable to injection attacks due to improper sanitization of user data.

True

False

Question 4

(2 marks)

EtterPeak is a widely used tool for network analysis in multiprotocol diverse networks.

True

False

Question 5

(2 marks)

The four primary security principles related to messages are Confidentiality, Access Control, Integrity, and Non-repudiation.

True

False

Long Form

Question 1

(10 marks)

Write a function that matches a string that has an a followed by zero or one 'b'.

Type your answer here...



Question 2

(10 marks)

Write a function to check whether the given string starts and ends with the same character or not using regex.

Type your answer here...



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